

Context-Aware Named Entity Recognition with Entity Linking and Explainability for Enhanced Natural Language Understanding

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by

Ritashree Mukherjee

under the guidance of

Prof. Payel Bag
Prof. Hasnahana Khatun



DEPARTMENT OF CSE – ARTIFICIAL INTELLIGENCE AND MACHINE
LEARNING

FUTURE INSTITUTE OF TECHNOLOGY

Garia, Boral, Kolkata – 700154

West Bengal

2025

Statement of Originality

I, **Ritashree Mukherjee**, University Roll No. **34230922025**, a student of **B.Tech (7th Semester)**, do hereby declare that the work entitled “**Context-Aware Named Entity Recognition with Entity Linking and Explainability for Enhanced Natural Language Understanding.**” submitted by me is an original work carried out under the prescribed academic framework as part of my undergraduate studies.

This work is the outcome of my or own own efforts, and to the best of my knowledge and belief, it has not been submitted previously to any university or institution for the award of any degree or diploma. All information, data, and materials obtained from other sources have been duly acknowledged and properly cited in accordance with academic norms and ethical conduct.

I further declare that this work complies with the applicable academic rules and regulations, and that I have taken due care to avoid plagiarism in any form.

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*This is to certify that the work contained in this thesis entitled “**Context-Aware Named Entity Recognition with Entity Linking and Explainability for Enhanced Natural Language Understanding**” is a bonafide work of **Individual Student Name (roll no)** carried out in the Department of Computer Science and Engineering–Artificial Intelligence and Machine Learning, Future Institute of Technology, Boral, Kolkata, under my supervision and that it has not been submitted elsewhere for a degree.*

Dr. Pradipta Kr. Banerjee
Professor & Head,
Department of CSE–AIML,
Future Institute of Technology,
Boral, Kolkata
December, 2025

Prof. Payel Bag
Prof. Hasnahana Khatun
Assistant/Associate Professors,
Department of CSE–AIML,
Future Institute of Technology,
Boral, Kolkata

External Examiner:

Designation:

Department:

College:

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PROGRAM OUTCOMES (POs)

1. **Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialization to develop the solution of complex engineering problems.
2. **Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
3. **Design / Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society, and environment as required.
4. **Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.
5. **Engineering Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
6. **The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

7. **Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.
8. **Individual and Collaborative Team Work:** Function effectively as an individual, and as a member or leader in diverse teams/multi-disciplinary teams.
9. **Communication:** Communicate effectively and inclusively within engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, considering cultural, language, and learning differences.
10. **Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
11. **Life-Long Learning:** Recognize the need for, and have the preparation and ability for (i) independent and life-long learning, (ii) adaptability to new and emerging technologies, and (iii) critical thinking in the broadest context of technological change.

Chapter 1

Introduction

Natural Language Understanding (NLU) is a core area of Artificial Intelligence that focuses on enabling machines to understand, interpret, and derive meaningful information from human language. One of the fundamental tasks in NLU is Named Entity Recognition (NER), which involves identifying and classifying important entities such as persons, locations, organizations, dates, and numerical values from unstructured text. NER plays a crucial role in many real-world applications, including information extraction, search engines, chatbots, question-answering systems

Traditional NER systems rely on rule-based methods or classical machine learning techniques that use hand-crafted features. While these approaches work well in simple scenarios, they often fail to handle complex sentence structures and ambiguous words. For example, the same word may represent different entities depending on the context, which traditional NER models are unable to accurately resolve.

To overcome these challenges, context-aware Named Entity Recognition has emerged as an advanced approach that uses deep learning and transformer-based models such as BERT. These models consider the surrounding words and sentence-level context while identifying entities, leading to improved accuracy and robustness.

1.1 **Overview of Named Entity Recognition**

Named Entity Recognition (NER) refers to the process of identifying named entities within a given dataset or corpus. These entities typically include person names, organizations, time expressions, locations, and more. However, the scope of named entities can extend to encompass entities from various domains, such as medical terms, food items, and even user-defined entities present within the corpus. For example, consider the sentence: "John bought 500 shares of Acme Corp. in 2016."

NER would annotate the entities as follows: "John" [Person] bought 500 shares of "Acme Corp." [Organization] in "2016" [Time]. In essence, NER serves as a vital component in natural language processing, facilitating the extraction and categorization of named entities, thereby enabling a deeper understanding of textual data.

Existing systems for Named Entity Recognition (NER) using Natural Language Processing (NLP) vary in complexity and effectiveness, with many leveraging state-of-the-art techniques and models to achieve accurate entity recognition.

1.2 **Importance of Context Awareness in NER**

Named Entity Recognition (NER) is a key task in Natural Language Processing that involves identifying and classifying entities such as persons, locations, organizations, and other meaningful elements from text. While traditional NER systems focus on individual words or limited local features, they often fail to understand the broader meaning of a sentence. This limitation makes context awareness an essential component in modern NER systems. As discussed by Matta et al. [MMH24], context awareness plays a crucial role in NLP tasks.

Context awareness enables NER models to consider the surrounding words, sentence structure, and overall meaning while identifying entities. Many words in natural language are ambiguous and can represent different entities depending on how they are used. For

example, the word “Apple” may refer to a fruit or a technology company, and the correct interpretation depends entirely on the context of the sentence. Context-aware NER models can accurately distinguish such cases by analyzing neighboring words and semantic relationships.

Another important advantage of context awareness is improved handling of complex and long sentences. Real-world text often contains nested entities, informal language, and varying grammatical structures. Context-aware models, especially those based on transformer architectures like BERT, capture long-range dependencies and sentence-level information, allowing them to recognize entities more accurately even in complex linguistic scenarios.

1.3 **Organization of The Report**

This chapter provides the background for the topics covered in this report. We begin with an overview of Named Entity Recognition (NER) and its significance in natural language processing. Next, we discuss the role of context-awareness in NER, which helps improve the accuracy of entity recognition in text. We also describe the dataset used for our experiments, CoNLL-2003, and its structure.

Further, we explain the concepts behind BERT-based NER models and their advantages over traditional approaches. The chapter also highlights the difference between rule-based, statistical, and deep learning-based methods for NER.

The rest of the report is organized as follows: Chapter 2 provides a review of prior works in the field of NER. Chapters 3 and 4 describe our proposed methodology and experimental results for improving context-aware NER using BERT-based models. Finally, Chapter 5 concludes the report and suggests directions for future work.

Chapter 2

Review of Prior Works

This chapter presents a survey of existing research and methodologies related to Named Entity Recognition (NER). It highlights the evolution of NER techniques from traditional rule-based to modern transformer-based models. Special emphasis is given to context-aware methods and the use of benchmark datasets that have contributed to advancements in NER performance. Early research in NER primarily relied on rule-based systems that used handcrafted linguistic rules and dictionaries. With the advancement of deep learning, models were able to automatically learn feature representations and capture sequential context, leading to improved performance compared to traditional statistical methods. More recently, transformer-based models have significantly transformed the NER landscape. Bidirectional Encoder Representations from Transformers (BERT) introduced a self-attention mechanism that enables bidirectional context modeling, allowing the model to understand the semantic meaning of words based on their surrounding context. Several studies have demonstrated that fine-tuning pretrained BERT models on benchmark datasets such as CoNLL-2003 achieves state-of-the-art results in NER tasks.[VSP⁺17]

2.1 Literature Review

Table 2.1 gives the overall literature review of the project. Natural Language Processing has evolved significantly over the years [nlp19].

Concept (Sequential)	Brief Description
Named Entity Recognition (NER)	Identifies and classifies entities such as Person, Location, Organization, and other predefined categories in text.
Context Awareness	Utilizes surrounding sentence-level context to enhance entity classification accuracy and resolve ambiguities.
Dataset: CoNLL-2003	A widely used benchmark dataset containing annotated named entities for training and evaluating NER models.
Text Preprocessing	Involves sentence segmentation and BIO tagging to structure textual data for effective model learning.
Tokenization	Uses BERT WordPiece tokenization while aligning subword units with corresponding NER labels.
BERT-Based NER	Fine-tunes a pretrained BERT model on the CoNLL-2003 dataset to perform context-aware token-level classification.

Table 2.1 Literature Review: Evolution of Named Entity Recognition

2.2 Conclusion

The literature review highlights the evolution of Named Entity Recognition (NER) techniques from traditional entity classification to context-aware models leveraging deep learning. Preprocessing methods, such as sentence segmentation and tokenization, play a critical role in structuring textual data for effective model training. Context-aware approaches, particularly those using BERT-based models, have significantly improved token-level classification accuracy by utilizing surrounding sentence context and resolving ambiguities. Benchmark datasets like CoNLL-2003 provide standardized evaluation frameworks, enabling con-

sistent performance comparison across models. Overall, integrating context awareness and advanced tokenization strategies has proven essential for enhancing NER systems, laying the foundation for explainable and entity-linked models in natural language understanding.

Chapter 3

Motivation and Plan of Work

The primary motivation of this work is to design an accurate and robust system for Named Entity Recognition (NER) that not only identifies entities correctly but also links them to appropriate knowledge bases for enhanced natural language understanding. In complex texts, many traditional NER systems fail to capture contextual nuances and semantic relationships, often resulting in ambiguous or incomplete entity recognition. Moreover, the absence of entity linking limits the ability to connect extracted information to structured knowledge, which is crucial for applications such as information retrieval, question answering, and knowledge graph construction. Therefore, our approach emphasizes context-aware modeling combined with entity linking, aiming to improve both the precision and practical utility of NER systems in diverse real-world scenarios.

The plan of work is executed in several stages. First, the textual dataset undergoes preprocessing, including tokenization, part-of-speech tagging, normalization, and handling of entity variations and ambiguous terms. Next, context-aware NER models are trained using transformer-based architectures such as BERT, RoBERTa, or similar models that capture both local and global contextual information in the text. Following entity recognition, an entity linking module is applied to map recognized entities to external structured

knowledge bases such as Wikidata or DBpedia, resolving ambiguities and enriching the semantic representation of the text. Recent advances using deep learning approaches have been discussed in detail in [nlp20].

3.1 **Proposed Methodology**

A. Input Text

The input to the proposed NER system consists of raw textual sentences obtained from the CoNLL-2003 dataset. The dataset contains annotated news articles in both English and German languages and is widely used as a benchmark for Named Entity Recognition tasks.

The sentences originate from real-world news sources and include naturally occurring named entities such as persons, organizations, locations, and miscellaneous entities. Each sentence is composed of sequential tokens, where each token is assigned a corresponding entity label following the BIO (Beginning, Inside, Outside) tagging scheme.

Before model training, the raw input text undergoes initial preprocessing to ensure compatibility with the transformer-based architecture. This includes sentence segmentation and preparation of token-level annotations. Using real-world textual data allows the model to learn meaningful contextual representations, which are essential for accurate entity recognition.

B. Text Preprocessing

Text preprocessing is performed to convert raw textual data into a structured format suitable for model training. This step includes sentence segmentation and sequence tagging using the BIO (Beginning, Inside, Outside) labeling scheme. Noise such as irregular spacing and irrelevant symbols is removed to ensure data consistency.

As a result of preprocessing, model-ready entity lists are created in the form of tokenized sentences and their corresponding NER labels. These structured inputs enable efficient learning by the transformer-based model.

C. Tokenization

In this step, the preprocessed sentences are tokenized using the BERT WordPiece tokenizer. WordPiece tokenization breaks words into subword units, allowing the model to handle unknown and rare words effectively.

The original NER labels are carefully aligned with the generated subword tokens to maintain correct entity boundaries. The final output of this step is prepared in a PyTorch-compatible format, which serves as direct input to the BERT-based NER model.

D. Context-Aware Named Entity Recognition

A pretrained BERT model is fine-tuned to perform the Named Entity Recognition task. BERT captures both left and right contextual information for each token through its bidirectional self-attention mechanism.

This context-aware labeling enables accurate token-level classification, particularly in cases where entity meaning depends on surrounding words. Fine-tuning allows the model to adapt pretrained linguistic knowledge to the specific NER task.

E. Training and Testing Data

The CoNLL-2003 dataset is utilized for training, validation, and testing of the proposed NER system. The training dataset is used to fine-tune the BERT-based model, allowing it to learn task-specific patterns.

A separate validation dataset monitors model performance during training and helps prevent overfitting. Finally, the testing dataset evaluates the overall accuracy and generalization capability of the trained model on unseen data. Context-aware approaches have shown improved performance in jointly modeling named entity recognition and relation extraction tasks [YJB21].

3.2 Conclusion

In this work, we proposed a context-aware Named Entity Recognition (NER) system based on a fine-tuned Bidirectional Encoder Representations from Transformers (BERT) model to accurately identify named entities in real-world textual data. The system was developed and evaluated using the CoNLL-2003 dataset, which contains annotated news articles in English and German and represents a standard benchmark for NER research. The overall methodology places strong emphasis on systematic text preprocessing to ensure high-quality and well-structured inputs for the learning process.

The preprocessing stage involves sentence segmentation, BIO-based sequence tagging, and noise removal, which together transform raw text into a clean and consistent format suitable for model training. This step is crucial for preserving correct entity boundaries and maintaining reliable token-label correspondence. Tokenization is performed using the BERT WordPiece tokenizer, which effectively handles rare, compound, and previously unseen words by breaking them into meaningful subword units.

By fine-tuning the pretrained BERT model on the NER task, the system is able to capture both left and right contextual information for each token through BERT’s bidirectional self-attention mechanism. This contextual understanding enables precise token-level classification, particularly in cases where entity interpretation depends heavily on surrounding words. The model is trained, validated, and tested under a consistent experimental setup to ensure robustness, stability, and strong generalization to unseen text.

Overall, the proposed framework demonstrates the effectiveness of context-aware transformer-based models for Named Entity Recognition. By combining rigorous preprocessing, subword-aware tokenization, and bidirectional contextual learning, the system provides a reliable and efficient solution for extracting meaningful entities from complex natural language text.

Chapter 4

Implementation and Experimental Results

The proposed context-aware NER system uses a fine-tuned BERT model on the CoNLL-2003 dataset to accurately identify entities in English and German news text. The text is preprocessed with sentence segmentation, BIO tagging, and noise removal, then tokenized using BERT WordPiece to handle rare or unknown words. The model captures bidirectional context, enabling precise token-level classification. Experimental evaluation on training, validation, and testing sets shows that the system reliably recognizes entities, preserves correct boundaries, and demonstrates strong overall performance, providing a robust framework for enhanced natural language understanding.

4.1 Implementation

The implementation of the Context-based Named Entity Recognition (NER) model was carried out using the BERT-base-cased architecture from the HuggingFace Transformers library, which provides pre-trained contextual embeddings capable of capturing semantic and syntactic nuances from text. Tokenization was performed using BertTokenizerFast, a tokenizer that efficiently splits input words into subword units while preserving the mapping

between original words and subwords. This ensures that the token-level labels, such as BIO tags, remain correctly aligned with the tokenized input, which is critical for accurate NER predictions.

To prepare the dataset for training, a custom `NERDataset` class was implemented. This class converts raw sentences and their corresponding BIO labels into the format required by BERT, including input IDs, attention masks, and label IDs. Special care was taken to handle padding tokens and subword tokens: padding tokens were ignored during loss computation, and subword tokens resulting from word-piece tokenization were also masked to avoid assigning multiple labels to parts of the same word. This preprocessing step ensures that the model focuses only on meaningful tokens while maintaining alignment between the input and target labels.

For experimentation, a subset of the CoNLL-2003 dataset was used, consisting of 12,000 sentences for training and 3,000 sentences for validation. The model was fine-tuned using the HuggingFace `TrainingArguments` API, with training configured for 6 epochs, a batch size of 8, and a learning rate of $3e-5$. Evaluation of the model was performed at the end of each epoch to monitor performance on the validation set, and checkpoints of the trained model were saved after each evaluation step. This setup allowed for systematic monitoring of the training process, helping prevent overfitting while ensuring the model progressively learned to identify entities in context.

Component	Description
Tokenizer	<code>BertTokenizerFast</code> from BERT-base-cased
Dataset Class	Custom <code>NERDataset</code> for token-label alignment
Training Data	12,000 sentences from CoNLL-2003
Validation Data	3,000 sentences from CoNLL-2003
Model	BERT-base-cased for token classification
Training Epochs	6

Table 4.1 Implementation Summary of Context-based NER

4.2 Improved Method

Although the current Context-based NER model demonstrates strong performance, further advancements include the integration of Entity Linking (EL) to associate recognized entities with unique identifiers, as well as the addition of Explainability mechanisms to provide insights into the model’s predictions and decision-making process.

4.3 Detail of Dataset - CoNLL-2003

CoNLL-2003 Dataset There are eight items total within CoNLL 2003 NE data (Tjong Kim Sang and De Meulder, 2003), and they include two different languages: German and English. The dataset focuses on several categories of named entities: people, places, organizations, and names of other entities that fail to fit into the former categories. Each of these languages has a development file, a training file, and a testing file, along with big, unannotated data. The functionality of the former three files is the same as in the CoNLL-2002 dataset. However, the difficult part of this collaborative work was finding a means to use the unannotated data within the learning process.

4.4 Detail Experimental Results

In this study, a context-aware Named Entity Recognition (NER) system was implemented using a BERT-based architecture to accurately identify entities in text. The CoNLL-2003 dataset, containing English and German news articles with annotated entities such as persons, organizations, locations, and miscellaneous categories, was used. Raw text underwent preprocessing including sentence segmentation, BIO tagging, and noise removal, producing structured inputs of tokenized sentences with aligned entity labels.

The tokenized data was processed using the BERT WordPiece tokenizer, with careful alignment of original NER labels to subword tokens to preserve correct entity boundaries. A pretrained BERT model was then fine-tuned, enabling the system to leverage bidirectional

contextual information for precise token-level classification.

The dataset was split into training, validation, and testing sets. The model was fine-tuned on the training set, validated during training, and evaluated on the test set for generalization. Experimental results show that the model effectively merges subword tokens into complete entities, accurately recognizing entities and maintaining correct boundaries even for complex or rare words . Overall, the evaluation confirms that the BERT-based, context-aware NER system provides robust performance and reliable entity recognition for real-world applications.

Raw output: [('El', 'B-PER'), ('on', 'I-PER'), ('Mu', 'I-PER'), ('sk', 'I-PER'), ('Founded', 'O'), ('Space', 'B-ORG'), ('x', 'I-ORG')]
Merged output: [('Elon', 'B-PER'), ('Musk', 'I-PER'), ('Founded', 'O'), ('Spacex', 'B-ORG')]

4.5 Conclusion

In this chapter, we presented the implementation and experimental evaluation of the proposed context-aware Named Entity Recognition (NER) system using a fine-tuned BERT-base-cased model. The system was trained and validated on the CoNLL-2003 dataset, covering both English and German news text. Through preprocessing, BIO tagging, and BERT WordPiece tokenization, the model was able to handle rare or unknown words while maintaining correct token-level label alignment.

The experimental results demonstrate that the proposed approach reliably identifies named entities, preserves accurate entity boundaries, and performs well across diverse textual contexts. The model effectively leverages bidirectional context for precise classification, confirming the robustness of the system for practical natural language understanding tasks.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This thesis investigated the application of context-aware machine learning techniques for Named Entity Recognition (NER) with the objective of developing an accurate, reliable, and context-sensitive system for identifying named entities in natural language text. The study was systematically structured to address the essential stages of the NER pipeline, including data preprocessing, tokenization, model selection and training, performance analysis, and contextual understanding of model behavior. This structured methodology ensured clarity, reproducibility, and experimental robustness throughout the research.

During the preprocessing phase, significant effort was devoted to ensuring data quality, consistency, and proper label alignment. The widely used CoNLL-2003 dataset was carefully prepared through sentence segmentation, BIO tagging, and noise handling to generate clean and structured inputs suitable for training a deep learning model. Particular attention was paid to preserving entity boundaries and maintaining correct token-label correspondence. The use of BERT WordPiece tokenization enabled effective handling of rare and unseen words while retaining meaningful semantic information, which is especially important in real-world textual data.

Several modeling strategies were explored, with a strong emphasis on transformer-based

architectures due to their ability to capture long-range contextual dependencies. A pre-trained BERT-based NER model was fine-tuned on the processed dataset, allowing it to leverage bidirectional self-attention to understand both preceding and following context for each token. This context-aware learning capability enabled the model to disambiguate entities more effectively, particularly in complex and linguistically rich sentences where traditional models often struggle.

Beyond overall system performance, this thesis emphasized a conceptual understanding of context-aware representations and their impact on named entity recognition. By analyzing how bidirectional attention and subword-level representations contribute to improved learning, the study highlights the strengths of transformer-based approaches for NER tasks. The findings demonstrate that incorporating contextual awareness.

Overall, this work confirms that context-aware NER systems provide a robust and scalable solution for natural language understanding. The proposed approach establishes a strong foundation for advanced NLP applications, including information extraction, text analytics, and intelligent text processing systems, and offers valuable insights into the practical benefits of transformer-based models for entity recognition tasks.

5.2 **Future Work**

In future work, the performance and capabilities of the current NER model could be significantly enhanced by incorporating entity linking (EL). By associating recognized entities with unique identifiers in structured knowledge bases, the model would gain the ability to understand entities in a broader context, resolve ambiguities, and improve overall semantic understanding. This enhancement would also enable downstream applications such as question answering, recommendation systems, and information retrieval to leverage more accurate entity connections.

Additionally, the accuracy of the model could be further improved by training on larger and more diverse datasets. Incorporating datasets from different domains and sources

would expose the model to a wider variety of entity types and contexts, making it more robust in handling rare, ambiguous, or domain-specific entities. This would ensure that the model generalizes better to unseen data and maintains high performance in real-world scenarios.

Another important avenue for improvement is extending the system to support multiple languages. Multilingual NER capability would broaden the applicability of the model to global text sources, allowing it to process and understand content across various languages, scripts, and cultural contexts. This would be particularly valuable for international organizations, multilingual content platforms, and cross-border applications.

Finally, making the system real-time and user-friendly would greatly enhance its practical utility. By optimizing inference speed and designing an accessible interface, users could receive instant entity recognition results on input text, facilitating seamless integration into web applications, chatbots, or enterprise tools. Ensuring an intuitive and responsive system would not only improve user experience but also increase adoption and applicability in diverse real-world workflows.

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